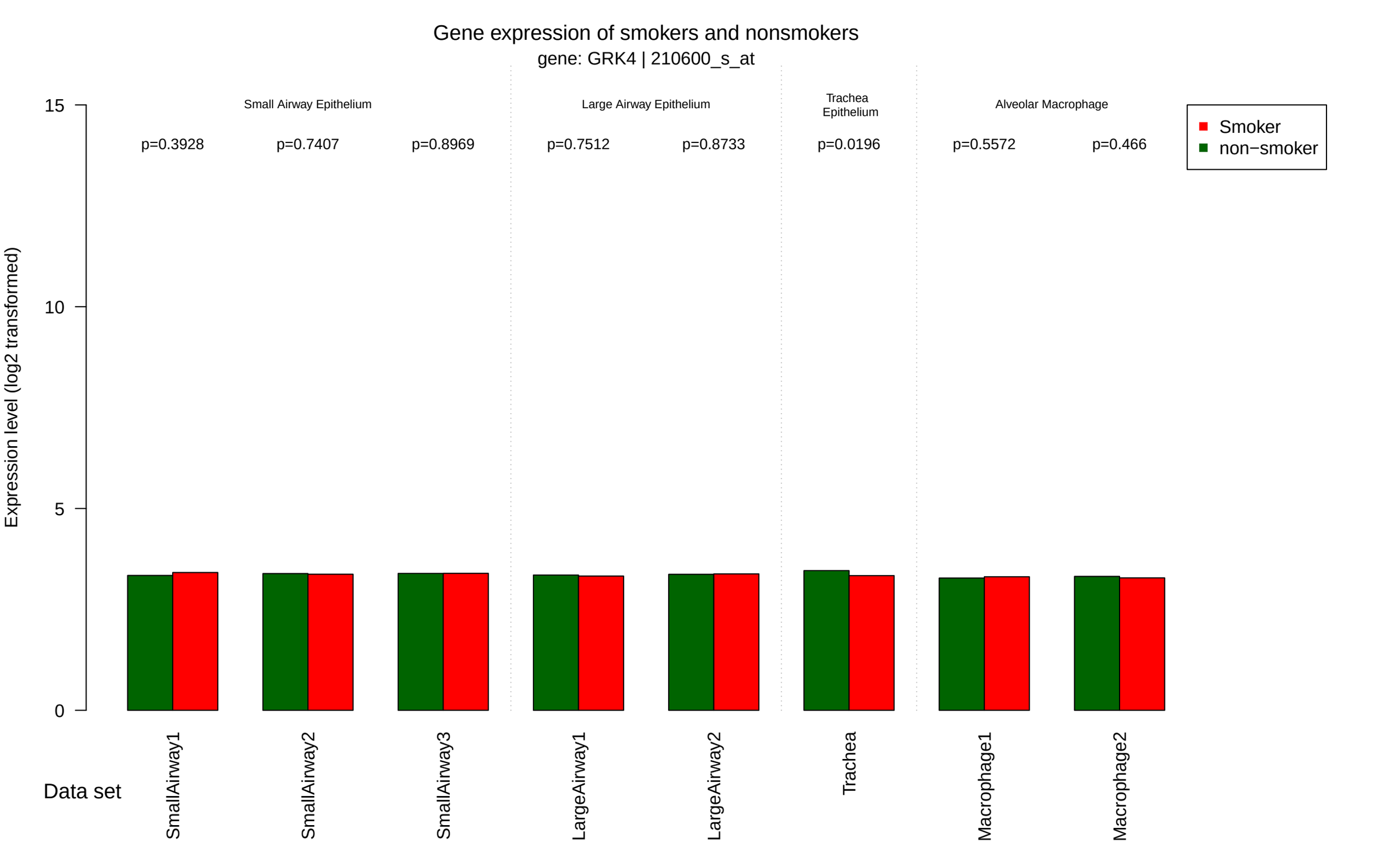
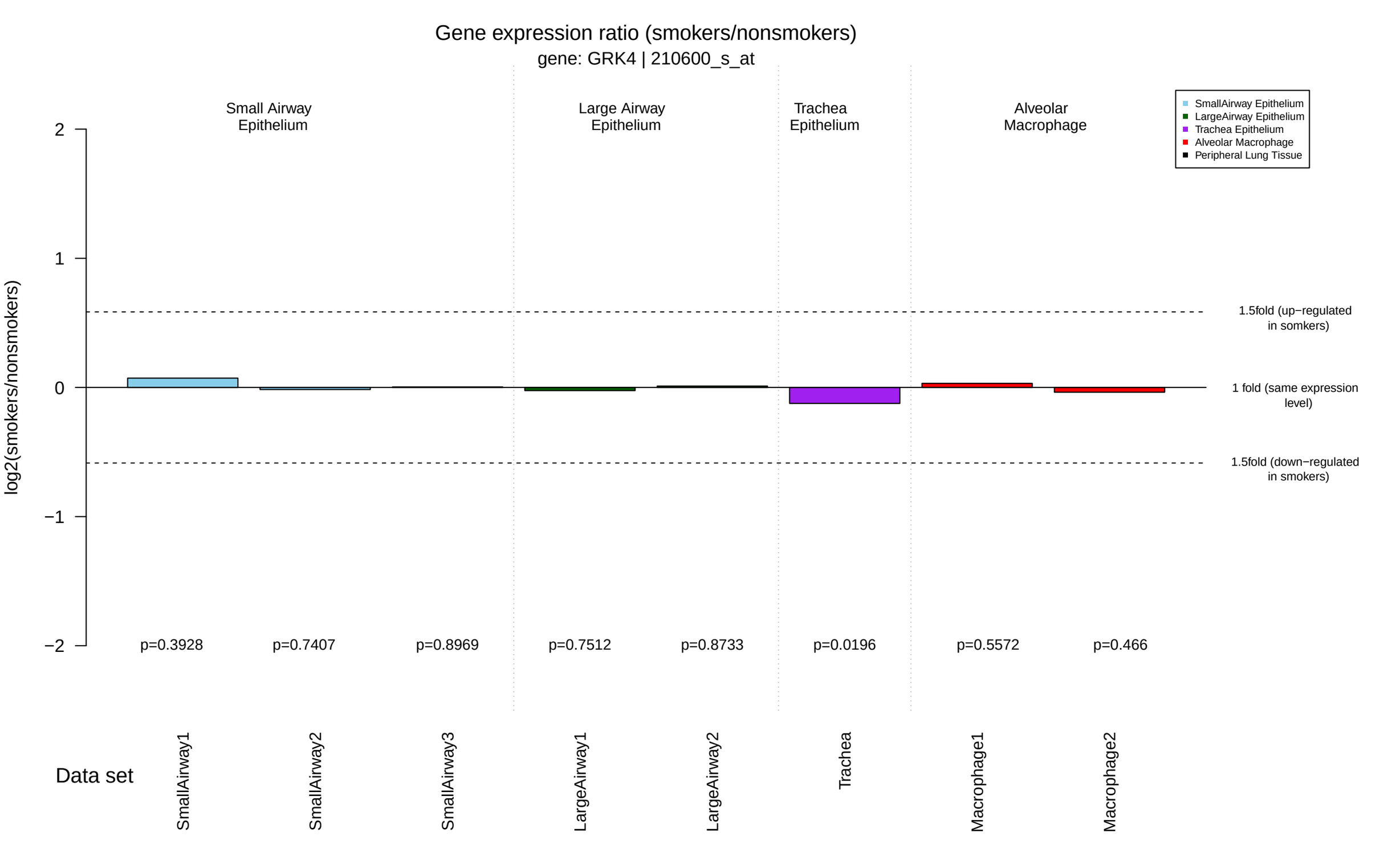
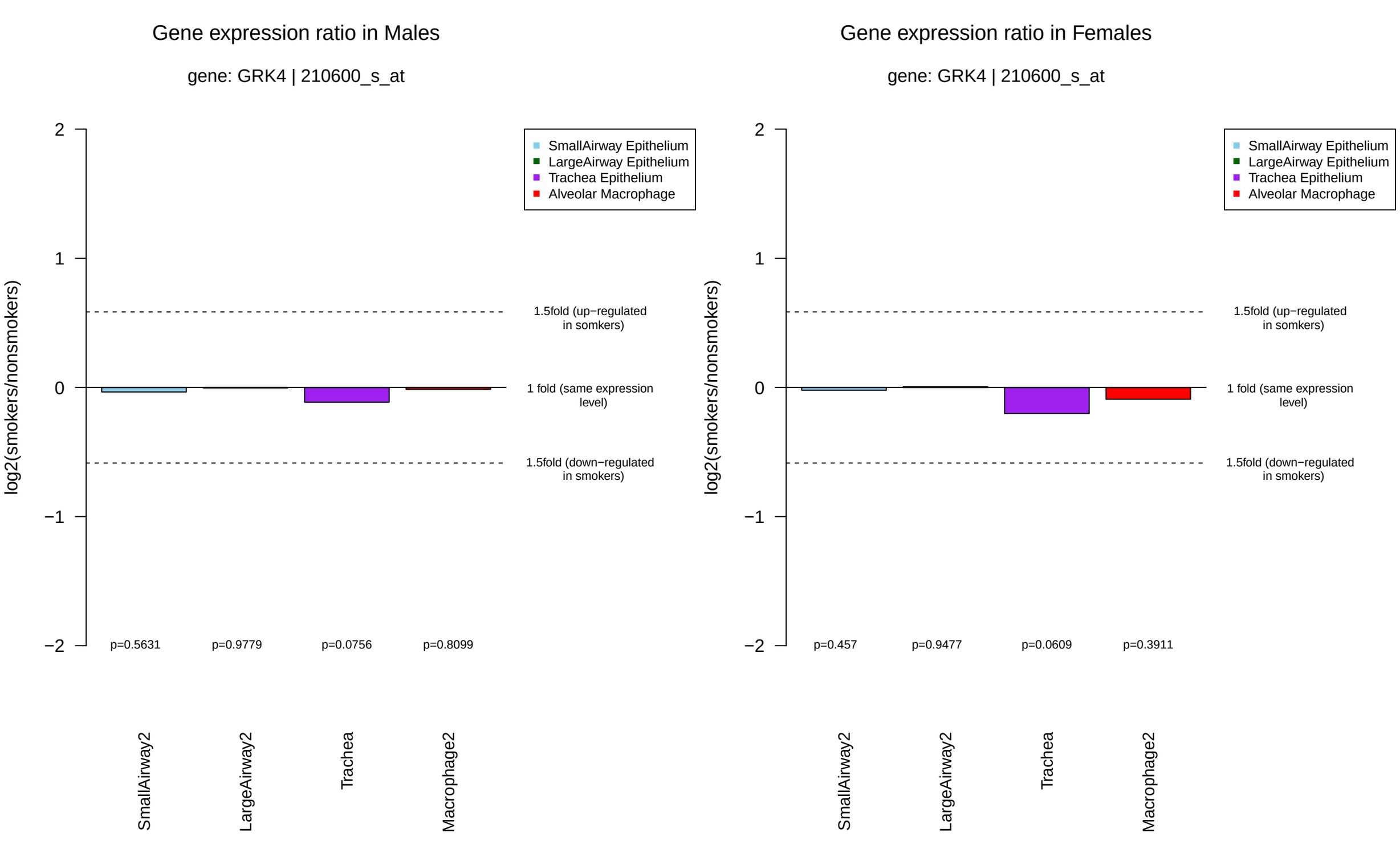
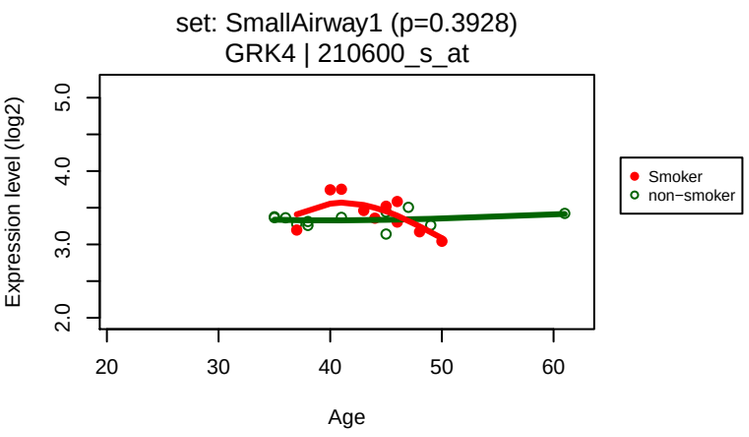
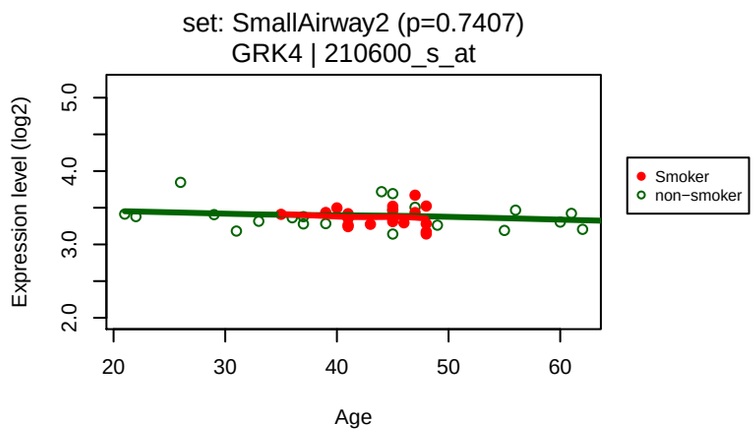
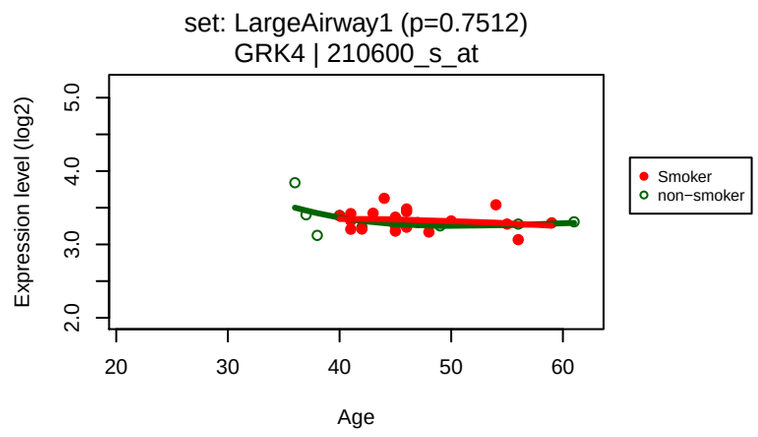

Gene expression of smokers and nonsmokers
gene: GRK4 | 210600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3928
p=0.7407
p=0.8969
p=0.7512
p=0.8733
p=0.0196
p=0.5572
p=0.466
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

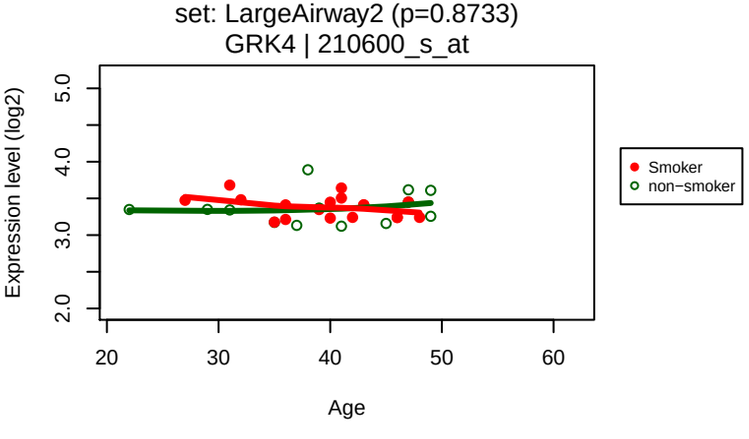

Gene expression ratio (smokers/nonsmokers)
gene: GRK4 | 210600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3928
p=0.7407
p=0.8969
p=0.7512
p=0.8733
p=0.0196
p=0.5572
p=0.466
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

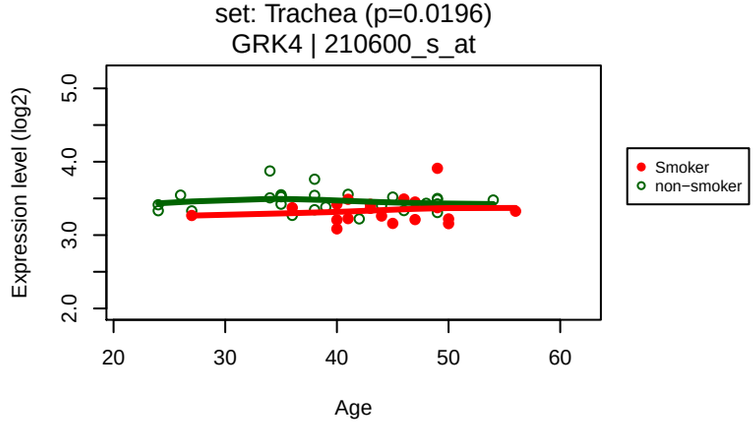

Gene expression ratio in Males
gene: GRK4 | 210600_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5631
p=0.9779
p=0.0756
p=0.8099
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK4 | 210600_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.457
p=0.9477
p=0.0609
p=0.3911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

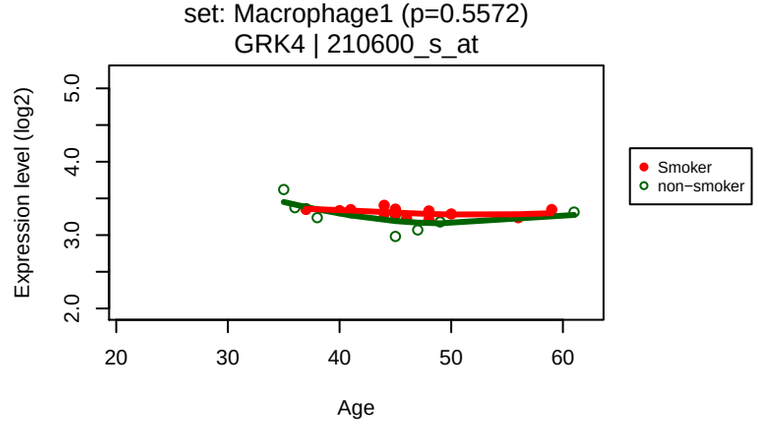

set: SmallAirway1 (p=0.3928)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker

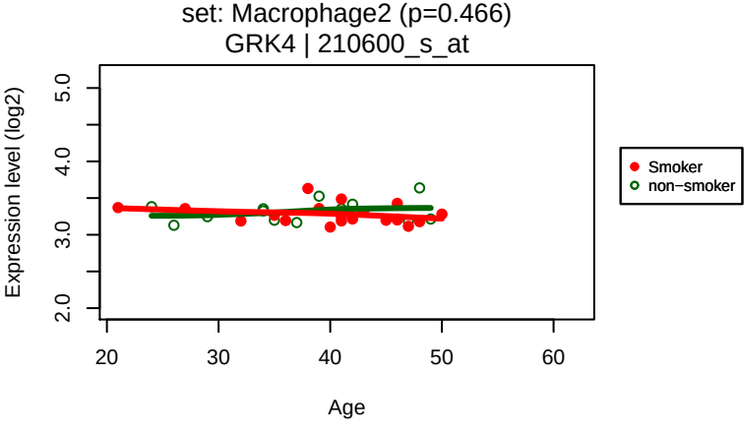

set: SmallAirway2 (p=0.7407)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker

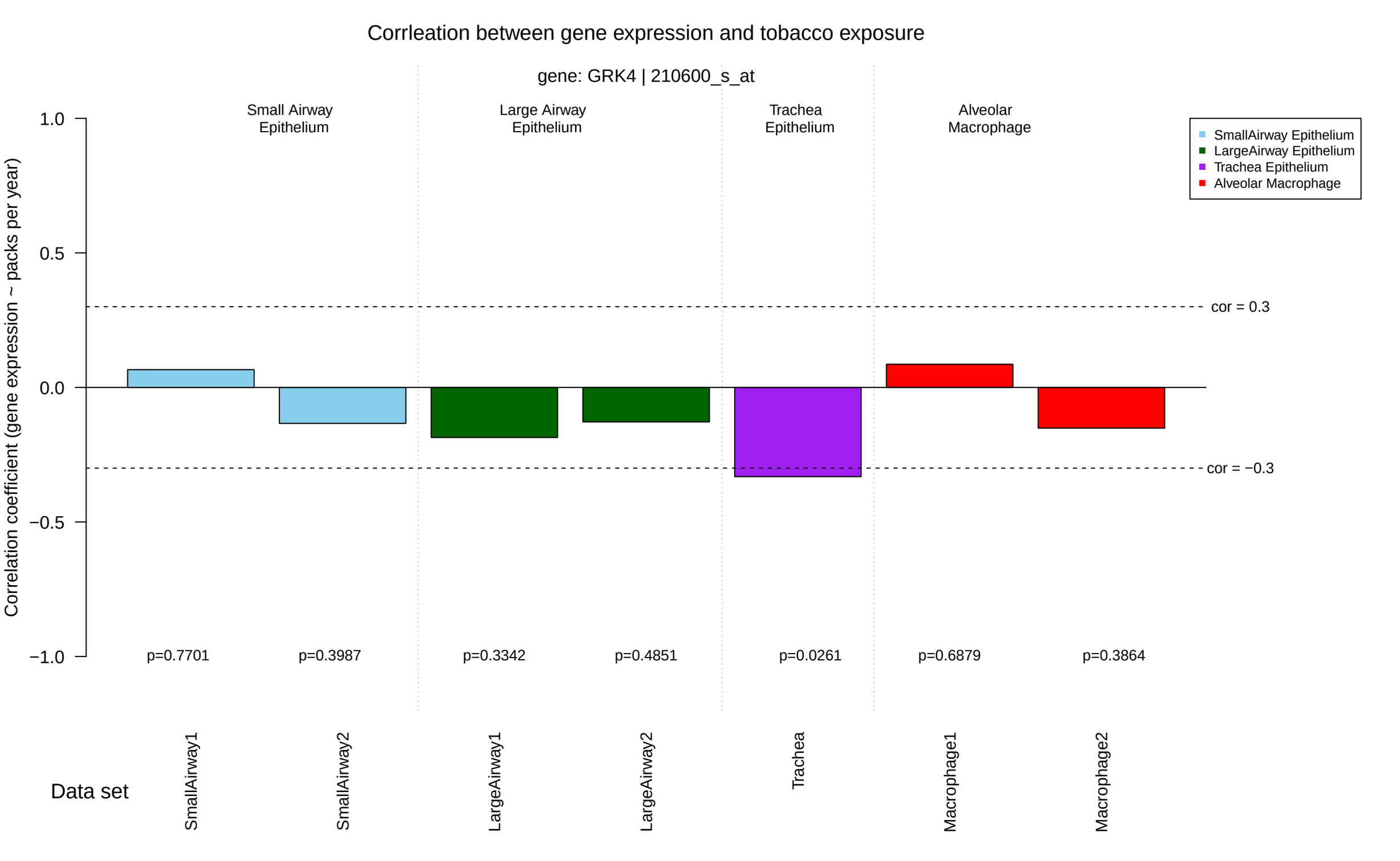

set: LargeAirway1 (p=0.7512)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8733)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0196)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5572)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.466)
GRK4 | 210600_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK4 | 210600_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7701
p=0.3987
p=0.3342
p=0.4851
p=0.0261
p=0.6879
p=0.3864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2